

How Effective are Enforcement Efforts Targeting Illegal Drugs?

Carlos Dobkin, Nancy Nicosia, and Matthew Weinberg*

January 2, 2013

Abstract

Although enforcement efforts are the primary approach to reducing illegal drug use and its harms in the U.S., the evidence on their effectiveness is mixed. This is due to the inherent difficulties in studying illegal activities and the lack of abrupt spatial and temporal variation in most enforcement efforts, both of which make it difficult to implement a clean research design. We overcome these challenges by using rich administrative records from law enforcement, public health, and workplace safety sources and by exploiting the staggered implementation of state laws targeting over-the-counter medicines that can be used to produce methamphetamine. We estimate that the regulations reduced the number of methamphetamine laboratories operating in a state by 50%, but find no significant change in methamphetamine consumption.

*The authors thank the Drug Enforcement Agency and the Substance Abuse and Mental Health Services Administration for providing data. We would also like to thank Joshua Angrist, Daniel Hosken, Diane Whitmore-Schanzenbach, Mark Stehr, and seminar participants at MIT, University of California at Berkeley, and University of California at Santa Cruz for helpful suggestions. Dobkin and Nicosia would like to acknowledge generous funding from the Substance Abuse Prevention Research Program at the Robert Wood Johnson Foundation. Finally, we appreciate the careful research assistance provided by Kristin Lang. Any mistakes are our own. Dobkin: UC Santa Cruz and NBER, cdobkin@ucsc.edu. Nicosia: RAND, nancy_nicosia@rand.org. Weinberg: Drexel University, mcw325@drexel.edu.

1 Introduction

The most recent estimates indicate that the economic cost of drug abuse in the U.S. exceeds \$200 billion annually (ONDCP 2004). The majority of these costs are due to lost productivity (71%) with the remainder due to spending on the criminal justice system (20%) and health care 9%.¹ Government efforts to reduce the substantial costs to society of drug abuse fall into three categories: prevention, treatment, and enforcement. In fiscal year 2010, approximately \$1.5 billion, \$3.7 billion, and \$9.8 billion were allocated to these three areas, respectively (ONDCP 2010). Despite the fact that enforcement efforts are the primary approach to reducing drug use and its harms in the U.S., there is little consensus on the efficacy of typical enforcement efforts. Some studies find no impact of enforcement on prices (DiNardo 1993), others document a negative relationship (Yuan and Caulkins 1998), and still others a positive relationship (Miron 2003). This is likely due to the difficulty in implementing clean research designs when studying the effects of enforcement efforts against illegal drugs.

The evidence is somewhat more compelling for very large enforcement efforts. Several recent studies that have examined abrupt, plausibly exogenous, changes in enforcement efforts have provided evidence that enforcement can reduce illegal drug use and drug-related harms. Several authors have documented that the reduction in heroin availability in Australia that resulted from the Taliban stamping out poppy production at least temporarily reduced some harms (Weatherburn, Jones, Freeman and Makkai 2002). Other authors have provided compelling evidence that the massive reduction in methamphetamine availability that occurred in the mid-1990s substantially (though temporarily) reduced drug related harms (Cunningham and Liu 2003), (Dobkin and Nicosia 2009). Their findings are perhaps not surprising given that the reductions in drug availability examined in these studies are likely the largest and most abrupt to have occurred in any developed country

¹A significant portion of the lost productivity is due to criminal activity as people who are incarcerated or pursuing criminal careers are typically not contributing significantly to the economy.

in the past 40 years.

Estimates of the effects of very large enforcement efforts may be of little practical value in predicting the effects of the enforcement efforts policymakers typically consider implementing. The reason is that most enforcement efforts at best moderately reduce the availability of the targeted drug. If modest reductions disproportionately impact the least addicted and criminally-involved users, then projections from these two large-scale interventions may be invalid. An additional concern about using the estimates from these two very large interventions is that they are identified largely from time-series variation and, consequently, provide limited insight into the persistence of enforcement efforts.

The objective of this study is to provide credible estimates of the impact of a more typical enforcement effort than the ones examined in Cunningham and Liu (2003), Dobkin and Nicosia (2009), and Weatherburn et al. (2002). We evaluate the impact of state restrictions on the sales of retail products that can be used to produce methamphetamine on the production and consumption of methamphetamine. Between 2004 and 2006, 35 states passed laws regulating the sale of over-the-counter (OTC) products that contain pseudoephedrine or ephedrine, the critical ingredients in most recipes for methamphetamine.² Focusing on this particular enforcement effort has the advantage that both the time period and geographical area affected by the enforcement are known and well-defined. To overcome the issues inherent in trying to observe the behavior of people engaged in illegal activity, we employ rich administrative datasets that are collected in the course of law enforcement, the provision of health services, and the establishment of workplace safety. These records are sufficiently rich that, unlike the survey records in the literature, it is possible to estimate rates of drug use and adverse events for each state on a monthly basis. Monthly estimates at the state level make it possible to take full advantage of the geographic and temporal variation in the implementation of these laws.

²States that did not implement their own legislation or implemented weak legislation became subject to the Combat Methamphetamine Enforcement Act (CMEA).

We examine the effect of the laws using a difference-in-difference model where the treatment is at the state level, with controls for state and month/year fixed effects and state-specific linear and quadratic time trends. In this model, identification requires that there are no state-level non-linear trends that are correlated with methamphetamine production and consumption and the introduction of the OTC restrictions. Our results are robust to adding controls for possible confounders and graphical analyses of event study models support the validity of our main conclusions. Event study models also provide insight into the dynamic effects of the OTC restrictions on methamphetamine production and consumption.

We document that implementing restrictions on the sales of OTC medicines significantly disrupts the production of methamphetamine within a state. We estimate that the laws cause a 50% reduction in the number of methamphetamine laboratories. While the reduction was largest for laboratories with limited production capacity, we estimate reductions for larger laboratories too. However, despite the disruption of domestic methamphetamine production, we find no evidence of an increase in the quality adjusted price. Next, we document that the laws resulted in no significant change in methamphetamine consumption. Because there are no survey measures of drug use with sufficient variation and sample size to support our analysis, we employ measures of consumption based on employee drug testing from Quest Diagnostics and hospital inpatient drug testing from the Health Care Utilization Project's Nationwide Inpatient Sample and State Inpatient Datasets. Although these estimates are specific to particular populations, they provide a good measure of drug use in the general population because the rates at which workers and hospital inpatients test positive for methamphetamine are strongly correlated with survey estimates of methamphetamine use in the general population. We were unable to find any evidence of a reduction in methamphetamine consumption using either of these two datasets. Finally, we examine the impact of the regulations on drug-related crime.

We find no evidence of a change in the arrest rate for possession or sales of addictive non-narcotics.

The remainder of the paper is organized as follows. Section II provides background on the production and physiological effects of methamphetamine and the efforts to reduce its availability. The administrative data sources used in this study are described in Section III. In the fourth section, we describe our empirical strategy. Section V presents our results. Finally, we conclude with a discussion of policy implications.

2 Background on Methamphetamine and Precursor Legislation

Methamphetamine is a central nervous system stimulant that induces a state of pleasure, high energy, and alertness that can last for up to 12 hours. The effects of methamphetamine are similar to those of cocaine but substantially longer lasting. Methamphetamine use and withdrawal are associated with a number of adverse physical and psychological events including cardiovascular complications, premature mortality, mood disorders, cognitive impairment, risk-taking and aggressive behaviors (NIDA (2002); Rawson, Huber, Brethen, Obert, Gulati, Shoptaw and Ling (2001); Simon, Domier, Sim, Richardson, Rawson and Ling (2001); Lynch, Kemp, Krenske, Conroy and Webster (2003); ONDCP (2006); SAMHSA (2008)). In 2005, methamphetamine use was associated with nearly 1000 deaths in the U.S. and more than 169,000 individuals sought treatment for methamphetamine or amphetamine.³ Some researchers and many law enforcement agencies have suggested that methamphetamine use is the cause of both property and violent crime (e.g., see Lynch, Kemp, Krenske, Conroy and Webster (2003); National Association

³Death counts are based on CDC tabulations of the Multiple Cause of Death file provided to the authors. Based on the Treatment Episode Dataset, methamphetamine accounts for the vast majority of drug treatment admissions for psychostimulants. See www.oas.samhsa.gov/2k8/methamphetamineTX/meth.pdf (Last accessed December 18, 2012).

of Counties 2005).

Manufacturing methamphetamine is a relatively simple, but potentially hazardous process that produces toxic byproduct damaging to the environment. Unlike most other illicit substances, methamphetamine is a synthetic product whose critical precursors-ephedrine and pseudoephedrine - require a complex large scale industrial process to manufacture. Because of the concentration in the production of ephedrine and pseudoephedrine, legislation aimed at reducing their diversion to methamphetamine production has the potential to reduce methamphetamine availability. In the last 25 years, the federal government has passed several laws intended to reduce the diversion of ephedrine and pseudoephedrine to illegal drug labs. The first of these was the Chemical Diversion and Trafficking Act of 1988 (CDTA), which regulated ephedrine and pseudoephedrine in bulk powder form, but left processed forms unregulated.⁴ This was followed by the Domestic Chemical Diversion Control Act of 1993 which placed restrictions on OTC ephedrine products (e.g. tablets) and increased DEA oversight of suppliers. Then, the Methamphetamine Control Act of 1996 tightened regulations on the sale of products containing methamphetamine precursors over 24 grams, but contained an exception for “blister packs”.⁵ Shortly thereafter, the Methamphetamine Anti-Proliferation Act of 2000 lowered the thresholds from 25 to 9 grams, but blister packs continued to be exempt. Each of these federal efforts caused methamphetamine producers to switch to other unregulated sources of precursors. Among these innovations was exploiting legitimate OTC consumer medicines containing precursors to support domestic production of methamphetamine (CRS 2007). Because many domestic labs turned to these OTC consumer products containing precursors, states began instituting controls on OTC cold and sinus medications such as Sudafed and Tylenol Cold in 2004. Federal legislation followed in the Combat Methamphetamine Act (CMEA)

⁴Other inputs were also regulated by these and other regulations, but we focus here on those that impact ephedrine and pseudoephedrine.

⁵The act also imposed monthly reporting requirements for “mail order” firms dealing in precursor chemicals.

of 2005.

The state and federal restrictions on OTC medicines containing methamphetamine precursors are the focus of our study. As shown in the last column of Table 1, thirty-five states passed laws restricting the sales of OTC medicines beginning in January 2004. The remaining fifty states and the District of Columbia became subject to the regulations in April 2006 when the federal laws regulating the sale of OTC medicines went into effect.

The most common state restrictions involved sales limits and product placement. Thirty-five states imposed limits on the sale of these products and required that they be placed out of customers' reach behind the counter or in a locked cabinet so as to deter theft. The majority of states also required that purchasers present identification and twenty-four states required that retailers maintain a logbook in order to prevent people from subverting the laws by making repeated purchases. We coded when each of these four restrictions went into effect in each state.⁶ Table 1 presents the correlation between these various laws within a state and month which reveals that they are highly correlated.

Figure 1 shows the time series of implementation for the four most common components of the laws. Purchase limits were typically an early component and were often bundled with restrictions intended either to deter theft or prevent people from working around the sales limits by making repeated purchases. In our main analysis, we focus on the earliest component—that is, the earliest date when each state implemented any of the four restrictions described above.

⁶We compared summaries of state legislation from the National Association of Chain Drug Stores, Office of National Drug Control Policy and National District Attorney's Association with the text of legislation from the 50 states and the District of Columbia. We resolved any inconsistencies by examining the text of the laws. In a few cases a Board of Pharmacy rule or executive order preceded the state and federal legislation.

3 Data

Because of the rapid rate of implementation across states, this study relies on several high-frequency administrative data sources to document the effects of the precursor regulations. We collected data on the number of methamphetamine labs, the price and purity of methamphetamine, positive drug tests for amphetamine among workers and among individuals admitted to the hospital, and drug related arrests. While many of the datasets were collected by state and federal agencies at the individual or transaction level, we use them to construct monthly measures for each state.

We have three measures that allow us to examine the impact of these regulations on drug production and markets, namely the number of labs identified by law enforcement, methamphetamine purity, and the nominal price of methamphetamine. The National Clandestine Lab Seizure System (NCLSS) contains information on labs seized or discovered by law enforcement.⁷ This dataset contains estimates of each site’s production capacity, which is a critical measure in determining whether OTC regulations disproportionately impacted smaller labs. We rely on the Drug Enforcement Agency’s System to Retrieve Information from Drug Evidence (STRIDE) to estimate the nominal prices and purity of methamphetamine.⁸ The STRIDE data have been criticized (Horowitz 2001), but remain the best source of information on the purity and price of illicit substances in the U.S. These data record the date, location, method of acquisition (e.g., seizure, purchase, etc.), purity of the drug as determined by DEA labs, and, in the case of purchases, the price. We construct our price and purity measures for large and small purchases using only observations on purchases in order to generate price and purity from a consistent

⁷A lab seizure is defined as “an illicit operation consisting of a sufficient combination of apparatus and chemicals that either has been or could be used in the manufacture or synthesis of controlled substances.” The NCLSS also records 1) “seizure of only chemicals, glassware, and/or equipment normally associated with the manufacturing of a controlled/illicit substance” and 2) locations “where discarded laboratory equipment, empty chemical containers, waste by-products, pseudoephedrine containers, etc., were abandoned/dumped.” See: http://www.michigan.gov/documents/EPIC143InstrNew_131649_7.doc (Last accessed December 29, 2012.)

⁸We have records from STRIDE and NCLSS for the period from January 2000 to March 2008.

sample.

Estimating what percent of the population consumes methamphetamine and how frequently they do so is complicated due to its illicit nature. Surveys such as the National Survey on Drug Use and Health (NSDUH) at best provide noisy annual measures of drug use that are of little use when leveraging changes in laws that occurred over just a couple of years. It is also likely that rates estimated from survey reports are biased downwards. For this reason, we measure methamphetamine consumption using drug testing data from Quest Diagnostics and the Health Care Utilization Project (HCUP). Quest Diagnostics is the largest provider of workplace drug tests. The rate of positive drug tests for amphetamine among employees tested by Quest Diagnostics is arguably a useful proxy for population wide drug use.⁹ Unfortunately, we only have data on positive drug tests among employees from January 2000 until April 2006 and so we are unable to follow late adopters for a full year.¹⁰ For a longer perspective on a larger sample of individuals, HCUP provides information on hospital admissions. These data allow us to identify hospital admissions (for any cause) that tested positive for amphetamine, which can be interpreted as a proxy for consumption. We rely primarily on HCUP’s Nationwide Inpatient Sample (NIS), which is a sample of hospital admissions for up to 37 participating states from January 2000 to December 2007. To increase the precision of our estimates, we supplemented the NIS data with the State Inpatient Dataset for January 2000 through December 2007 for the states of Arizona, New Jersey, Washington, and Iowa.¹¹

To examine the link between the OTC regulations and drug related arrests we use the Uniform Crime Reports (UCR). For the period from January 2000 to December 2007 we generate arrest rates for the possession and sale of illegal drugs for each state and month.

⁹We cannot differentiate amphetamines from methamphetamines in the drug testing and hospital data, but we do know from the TEDS data that more than 90% of the amphetamine abused in the U.S. is methamphetamine.

¹⁰These data were drawn from the ONDCP (2006) publication entitled “Pushing Back Against Meth”. We attempted to obtain additional state-level data directly from Quest Diagnostics to extend our analysis period, but were unsuccessful.

¹¹Data from Iowa were only available for January 2004 through December 2007.

The UCR is based on reports provided to the FBI by individual police agencies. While it is requested that agencies report monthly, many do not as reporting is voluntary. We drop observations from Alabama and Illinois because agencies in these states did not report at the monthly frequency.¹²

4 Empirical Strategy

Our empirical strategy exploits the variation in the timing of states' implementation of OTC regulations. Thirty-five states implemented retail and consumer regulations prior to the CMEA and the remaining states and the District of Columbia became subject to the federal CMEA in 2006. Figure 1 demonstrates when the states enacted various OTC restrictions. Because of the fairly rapid pace of implementation and because many states adopted laws in the middle of the year, the unit of observation is a state-month. We provide both graphical and regression based evidence.

We begin with a descriptive graphical analysis by plotting the average of each measure of methamphetamine production, consumption, and drug related crime over the sample period. These figures show national trends in methamphetamine production and consumption during the time period of our study. The figures also provide some insight into whether changes in the outcomes are roughly occurring during the time period when states were enacting the regulations.

After examining national trends in the outcomes over the sample period, we then implement a regression analysis. The regression analysis leverages the variation in when states enacted the OTC regulations to separately identify the effects of the regulations from common changes in outcomes across all states. For each outcome the regressions are

¹² In several other states there were large spikes across all arrest categories typically in June and December. These increases are likely due to a share of agencies reporting twice a year instead of monthly. In these cases, we replaced the outlier with the value in the preceding month within that state. Details are available upon request.

estimated using data beginning in January 2002. The end dates vary across data sources as described in Section III and in the notes at the bottom of the regression tables. We fit the following equation to the data using OLS:

$$Y_{st} = \beta OTC_{st} + \alpha_s + \gamma_t + \epsilon_{st} \quad (1)$$

where Y_{st} is the outcome in state s and time t . The independent variable of interest OTC_{st} is an indicator set to zero in months prior to implementation of the regulation, one afterwards, and the fraction of the month a regulation was implemented in months a regulation was enacted. Our main specification defines the OTC variable based on the earliest effective date for the four major retailer and consumer restrictions. We control for common time shocks to all states by including a fixed effect for each time period γ_t and we control for unobservable time-invariant differences across states by including a fixed effect for each state α_s . The coefficient of interest β measures the effect of the OTC regulation on the average outcome. When the outcome Y_{st} is a state average we weight the data to improve the efficiency of our estimates. We allow for serial correlation within a state and different variances in the error term across states by clustering our standard errors by state. We then expand on Model 1 by progressively adding state specific linear and quadratic time trends and time varying covariates that include the state unemployment rate, number of households receiving food stamps, average temperature and average precipitation.¹³

Before examining the results of estimating Model 1 we use event-study analysis to examine the timing of any change in each outcome. A key assumption necessary for specification 1 to estimate the effect of the regulation is that there are not unmeasured state specific trends correlated with the enactment of the regulations. The event study analysis allows us to examine any “pre-trend” that would raise concerns about our identification

¹³Previous research has shown that there is an effect of unemployment (Raphael and Winter-Ember 2001) and weather on crime (Jacob, Lefgren and Moretti 2007).

strategy and also allows us to trace out the dynamics of any policy effect. We implement the event study by estimating the following equation with OLS:

$$Y_{st} = \sum_{j=-10}^9 \pi_j 1(\tau_{st} = j) + \pi_{10+} 1(\tau_{st} \geq 10) + \alpha_s + \gamma_t + \delta_s * t + \omega_s * t^2 + \epsilon_{st} \quad (2)$$

where τ_{st} measures the month relative to the introduction of an OTC regulation, defined so that $\tau_{st} = 0$ if state s enacted an OTC regulation at any point in month t . As in the most flexible version of Model 1, we include state specific linear and quadratic time trends in order to mitigate the risk of OTC regulation endogeneity. We then produce graphs by plotting the estimated π_j coefficients with respect to event time, where the coefficients are measured relative to the month before an OTC restriction was put into place ($\tau_{st} = -1$).¹⁴

5 Results

In this section, we present our analysis of the impact of the laws regulating OTC medicines on the production of methamphetamine, its consumption, and drug-related crime. We begin by examining how the laws affected the number of discovered laboratories. We then document the extent to which the laws affected consumption as measured by drug testing. Finally, we examine whether the intervention reduced arrest rates for selling and possessing illegal drugs.

5.1 Methamphetamine Production

The intent of the OTC regulations is to prevent the diversion of over-the-counter medicines containing pseudoephedrine to the production of methamphetamine. Most recipes for methamphetamine require pseudoephedrine in combination with numerous other ingre-

¹⁴This was done by subtracting π_{-1} from each of the other event period coefficients.

dients.¹⁵ As discussed above, restricting access to over-the-counter sources of ephedrine and pseudoephedrine can affect the supply of methamphetamine in two ways. It can limit the number of producers with access to sufficient precursors to produce methamphetamine. Second, it can change the quality of available methamphetamine. Quality of methamphetamine will be reduced to the extent restrictions force producers to dilute their product or turn to inferior precursors. Alternatively, the quality of available methamphetamine may increase if the restrictions reshuffled production from lower quality producers to higher quality producers. We find evidence that limiting the availability of over-the-counter pseudoephedrine reduces the number of producers with access to sufficient precursors to produce methamphetamine. We find no evidence that the precursor regulations led to lower purity methamphetamine.

The ideal dataset for measuring the effect of the laws on methamphetamine production would be a census of labs for each state spanning a period before and after the laws went into effect. Given that methamphetamine producers have a strong incentive to hide their operations this is clearly infeasible. As an alternative, we examine data on the number of labs that come to the attention of law enforcement agents. The number of labs discovered by law enforcement agents each month is an unknown fraction of the total number of labs in operation that month. This detection probability is likely a function of a number of factors including (among other things) the effort on the part of the law enforcement agents, the likelihood of a lab catching fire, and reports from the public. The relationship between the percentage change in the number of discovered labs and percentage change in the number of labs in operation is given by the following equation:

$$\frac{D_{post} - D_{pre}}{D_{pre}} = \frac{p_{post} - p_{pre}}{p_{post}} \left(1 + \frac{Labs_{post} - Labs_{pre}}{Labs_{pre}} \right) + \frac{Labs_{post} - Labs_{pre}}{Labs_{pre}} \quad (3)$$

¹⁵Unlike ephedrine and pseudoephedrine, the other ingredients have numerous substitutes that have many legitimate uses and so are widely available.

where D is the number of labs detected, $Labs$ is the number of labs in operation, p is the probability of detection, and the subscripts denote whether the period under consideration is before or after the implementation of the OTC law. If there is not a large change in the detection probability when the OTC regulations are put in place, the percentage change in the number of discovered labs is a good estimate of the percentage change in the number of labs in operation. In fact, if $p_{post} = p_{pre}$, then the percent change in the number of labs discovered is identical to the percent change in the number of labs in operation. There is anecdotal evidence that the OTC laws may have slightly increased the probability that a given lab will be detected as some police departments found cause to visit residences of people whose names appeared repeatedly in OTC sales logbooks. If the probability of a lab detected increases with the law ($p_{post} > p_{pre}$), then using the percent change in the number of labs detected to estimate the percent change in the number of labs in operation will lead us to underestimate the reduction in labs by $100 * (\frac{p_{post} - p_{pre}}{p_{pre}})(1 + \frac{Labs_{post} - Labs_{pre}}{Labs_{pre}})$ percentage points.¹⁶

In Figure 2, we present the average number of labs discovered for each month in our data. The labs have been split into three groups based on an estimate of the production capacity measured as the amount of methamphetamine that could be produced in a single production cycle. Two facts emerge from the figure. First, for each group, the figure shows a clear decrease in the average number of labs discovered beginning in early 2004. Second, by 2007 the average number of labs discovered in a state had reached less than half of what it was before 2004 for each of the three groups of labs. Comparing Figure 1 to Figure 2 makes it clear that not all of the decline in the number of labs from 2004 to 2007 can be attributed to the OTC regulations. The decline began prior to the time period when states were enacting the sales restrictions. This motivates our regression analysis

¹⁶To get a sense of the possible scale of the bias consider the case where the OTC laws reduce the number of labs in operation by 50 percent and increase the probability of detection by 10 percent. Using the change in the number of labs detected will lead to an estimate of 45 percent which underestimates the change in the number of labs active by 5 percentage points.

that uses the variation in when states enacted the laws to estimate their impact.

Figure 3 presents the event study created by estimating Model 2. The dependent variable is the number of labs discovered in a state. The figure shows that the reduction in the number of labs was not entirely due to common reductions in labs discovered across all states. While there is no obvious trend in the outcome between 10 and 3 months prior to the implementation date, the decline appears to begin about two months before the laws went into effect. This is possibly due to the fact that some retailers came into compliance with the laws before the laws went into effect.

The corresponding regressions are presented in Table 2. Column 1 presents our most parsimonious model corresponding to equation 1, the second column adds state-specific linear time trends, the third column corresponds to the event study and adds quadratic state-specific time trends, and the final column adds the covariates. The findings from the event study are confirmed by the regressions. Across specifications, the laws led to a reduction in the number of labs discovered. Column 3 of the first panel indicates a decline in the total number of discovered labs of approximately 50 ($\frac{-7.02}{13.9}$) percent. The other three panels present the estimates of the decline broken out by the size of the lab. The reduction was similar across labs with capacity less than two ounces and labs with capacity between two and eight ounces at approximately 49 and 58 percent. For the largest labs the reduction was slightly smaller at 38 percent and not significant at the .05 level.

The OTC regulations clearly reduced the average number of discovered methamphetamine labs in a state. We next use estimates of the reduction in the number of labs from Model 1 and data on the distribution of lab sizes to estimate the effect of the regulations on a state's methamphetamine production capacity. Our data contains information on the number of labs discovered by state and month for labs grouped in six different production capacity intervals: labs with capacity less than two ounces, between two and eight ounces, eight ounces and one pound, two pounds and nine pounds,

ten pounds and nineteen pounds, and labs with capacity of at least twenty pounds. We estimate the total reduction in production capacity by first expanding on the results presented in Table 2 by estimating the reduction in the number of labs for all six lab size groups. The change in detected lab capacity is calculated by assigning the midpoint of the capacity range for each group and then taking the weighted average of the decrease in capacity lab sizes. Our estimate of the percent reduction in productive capacity is given by the following expression:

$$PercentCapacityReduction = \frac{\sum_{j=1}^6 s_j * \beta_j * Capacity_j}{\sum_{j=1}^6 s_j * E[DiscoveredLabs|PreOTC, Capacity_j] * Capacity_j} \quad (4)$$

where s_j is the fraction of labs that belonged to capacity group j before any OTC regulation was in place, β_j is an estimate of the reduction in the number of labs in group j , $E[DiscoveredLabs|PreOTC, Capacity_j]$ is the average number of discovered labs in group j conditional on no OTC restriction being in place, and $Capacity_j$ is the midpoint of the productive capacity interval for group j .¹⁷ If the detection probability is the same across labs of different sizes and there is not a large change in the detection probability when the OTC regulations are put in place, this is a good estimate of the percentage change in productive capacity.

We estimate that the OTC restrictions reduced a state’s capacity to produce methamphetamine by 50 percent. Our finding is robust to measuring capacity at the twenty-fifth or seventy-fifth percentile of each capacity interval. Doing so changes our estimate of the reduction in capacity by less than two percentage points.

We next estimate the impact of the OTC restrictions on the purity and price of methamphetamine. Prior work (Dobkin and Nicosia 2009) documented that a very large, sudden, and unexpected reduction in the availability of a precursor led to a temporary reduction in the purity of methamphetamine and an increase in its price.

¹⁷The shares for each of the six groups, ordered from smallest to largest, were .69, .23, .04, .02, .006, and .005. We evaluate the capacity of labs in the largest group at the left endpoint of twenty pounds.

Figure 4 documents changes in the average nominal price per gram and purity of methamphetamine over our sample period. We weight both series by the number of transactions used to compute the average outcome in each state and month. This reduces some of the volatility, though both series are still somewhat volatile. There is no clear shift in the nominal price per gram at any time between 2002 to 2008. While there is a large drop in the average purity of methamphetamine at the end of 2005 implying an increase in the real (purity adjusted) price of methamphetamine, there are other large changes in average purity prior to the time period when states were enacting the regulations.

Figure 5 presents the event study for price and purity and Table 3 displays estimates from the corresponding regressions. The figure shows an absence of pre-trends for both outcomes. The figure also shows no increase in the nominal price of methamphetamine. The purity measure if anything increases after the introduction of the OTC regulations. The regressions confirm the event study. The point estimates imply a change in the average price ranging from \$2.89 to \$ - 1.83, depending on specification. This is quite small relative to the pre-intervention mean of 49 dollars per gram. Across specifications, none of these estimates are significant at conventional levels. We estimate small increases in the purity of methamphetamine of between 7 and 10 percent of the pre-regulation average purity level, though the estimates are only significant at the 10 percent level in the three specifications that control for state-specific trends. Unsurprisingly, there is no evidence of changes in the price or purity of cocaine, crack, or heroin.

Overall, these results are compelling evidence that state OTC regulations disrupted local production of methamphetamine. The extent to which this disruption in methamphetamine production within a state affects consumption depends on how much of the consumed methamphetamine consumed in a state is locally-produced as opposed to other states or countries, eg Mexico, and how rapidly producers from other regions mobilized to meet the unfilled demand.

5.2 Methamphetamine Use

The reduction in the number of domestic labs may have increased the search costs of obtaining methamphetamine. In this section we analyze the impact of the OTC restrictions on two measures of methamphetamine consumption: positive workplace drug test and hospital toxicology test rates. We find no evidence that the laws reduced the rate of positive drug tests.

Documenting the effect of the OTC regulations on methamphetamine consumption is challenging because consumption is an illegal activity. Given that we implement a state-by-month panel approach, we would ideally have monthly measurements of what proportion of the population consumes methamphetamine on a typical day of the month and how much they are consuming. Surveys are of little use because of their limited sample sizes and infrequent nature (e.g. annual) and because the low prevalence of methamphetamine makes it difficult to generate precise month-by-state series. A second issue with survey data is that relying on self reports will probably result in consumption estimates that are substantially biased downward as there is substantial evidence that people underreport undesirable behavior.¹⁸

Fortunately, drug tests provide an alternative measure of methamphetamine consumption. The half-life of methamphetamine in the body is 9-12 hours and urine tests will typically return a positive test if the person has consumed methamphetamine in the prior 3 to 5 days. Though the detection window is sensitive to the amount consumed given the relatively short half life, even a 50 percent reduction in the amount of methamphetamine consumed will only reduce the period over which the use can be detected by 9-12 hours.

We have two sources of drug tests that we focus on in this study. The first are work-related drug tests conducted on employers' behalf by Quest Diagnostics. The workplace drug tests are not all from a random sample of workers. Although some of them are,

¹⁸?(?) compares survey reports of alcohol consumption with estimates of alcohol sales from tax records and documents that alcohol consumption in surveys is underreported by over 50 percent.

another portion of the tests are conducted for particular reasons such as pre-employment screens, accidents, or “for cause”. Tests that are done “for cause” are more likely to return a positive finding than tests that are conducted for other reasons. Not all employers test their employees. About 20 percent of employee drug tests in this series are from federally-mandated safety-sensitive workers while the remainder are from firms that have chosen to test their employees. The second source of drug test results are from hospital records. These are tests of people admitted as inpatients to the hospital where either the patient reported drug use or the clinicians treating them decided to test them based on the symptoms they presented. The downward bias that results from the fact that physicians choose to test only a subset of patients is likely to be modest as there is evidence that doctors do a good job of detecting drug use.¹⁹ A more significant issue with measures generated from the hospital data is that people who experience an inpatient hospital stay are not representative of the general population.

To obtain a sense of how well these two measures of drug use pick up variation in drug use, we compare them with each other as the correlation between them is due to their correlation with underlying use rates. The correlation between the two series is 0.77, suggesting that they are both pretty highly correlated with the underlying drug use in the general population.

We also compare the estimates from the workplace and hospital testing with estimates of state-level prevalence from the National Survey on Drug Use and Health during the period from 2004 to 2007. We are restricted to making the comparison based on state level means calculated from a four year period because the survey will not support an analysis over a shorter period.

We present these comparisons in Figure 6. Each graph is a scatter plot of the average percent of drug tests that were positive against the average fraction of survey respondents

¹⁹In a randomized trial ? (?) find that in an urban psychiatric emergency room only eight percent of people that had used drugs recently did not have a drug test ordered for them by the doctor treating them.

reporting they have used methamphetamine in the prior 12 months. The size of each bubble in the graphs is proportional to state population. The graphs reveals that the drug tests are highly correlated with survey measures of methamphetamine use. The rates at which employees and hospital patients test positive for amphetamine are highly correlated with survey estimates of methamphetamine use in the general population with (population weighted) correlations of 0.89 and 0.85, respectively.²⁰ The strength of these correlations suggest that these two drug testing measures are truly measuring underlying use. Despite the likely under-reporting in surveys, the percent of the population reporting use in the surveys is larger than the percent testing positive. This is likely due to the differing reference windows. The survey response is regarding the prior 12 months while the drug test will only pick up use in the prior 3-5 days.

We now turn to examining the impact of the OTC regulations on the rate at which individuals test positive for methamphetamine use in workplace drug tests. The workplace drug test data covers a shorter time period than our other datasets, with an endpoint of April 2006. Figure 5 shows that the share of employees that test positive for methamphetamine was slowly growing from 2002 through 2005. Figure 6 presents the results of the event study. While there is no obvious pre-trend in the conditional average positive test rate, there is also no clear change after the enactment of the OTC regulations. This is supported in each specification of our regressions, which are presented in Table 4. Each specification yielded a fairly precise estimate near zero. Column 3, which refers to the specification with the same controls as Figure 6, contains a point estimate of essentially zero. The standard error implies a 95 percent confidence interval with end-points that are less than ten percent of the average positive test rate prior when no OTC restrictions were in place. This is in stark contrast to the change observed for labs.

We also find no significant impact of the regulations on positive hospital toxicology

²⁰The state-specific estimates of methamphetamine use among the general population were generated by stacking four years of the National Survey on Drug Use and Health.

test rates, though our estimates are less precise than they were for the workplace drug tests. Figure 7 shows the time series of the average share of hospital admissions with a positive indication for methamphetamine in a state. We do not have a full time series of admissions data for a few of the states, so we include a line in the figure that plots the percent of the population included in the sample at each point in time. The average share of positive tests for methamphetamine at hospitals was generally increasing from January of 2002 through the end of 2005. The upwards trend broke sometime in the end of 2005 and by the end of 2007 the test rate was the same as it was at the beginning of 2002. Despite this pattern in the time series of average positive test rates, the event study graph presented in Figure 8 reveals no impact of the regulations on positive test rates. This implies that most of the changes in the raw averages of Figure 7 were driven by common changes in the test rates across all states. Table 5 presents the regression results. None of the point estimates are significantly different from zero at conventional levels. We also present results for opioids, cocaine, and marijuana. There is no significant increase in the test rates for any of these other drugs.

Despite the large reduction in the number of labs within a state after an OTC regulation is passed, there is no evidence that the laws led to a reduction in methamphetamine consumption. This suggests that a substantial portion of the methamphetamine consumed in a state is not produced within that state.

5.3 Drug Related Crime

Even though the OTC restrictions did not significantly reduce our measures of methamphetamine consumption, the restrictions may have had an effect on drug related crime given their large impact on local methamphetamine production. If the reduction in the number of labs led to fewer people selling methamphetamine, then we may observe a decline in the arrest rate for drug sales. In this section we study the impact of the OTC

restrictions on drug related crime. We find no evidence that the laws caused a change in arrest rates for the sale or possession of addictive non-narcotics.²¹

Figure 9 plots the average monthly arrest rates per 10000 person years for sales and possession of addicting non-narcotics, which includes methamphetamine, and addicting narcotics (eg heroin and other drugs with sleep-inducing properties). While the arrest rate for possession of both types of drugs increased over the beginning of our sample window, there is no clear trend or break in either average arrest rate for the sale of either type of drug.

Figure 10 presents the event study for drug related arrests. The event study reveals no clear change in any of the arrest rates after the law, but there is a systematic downward trend with an apparently constant slope over the entire event period for the arrest rate for the sale of addicting narcotics. The regressions are presented in Table 6. In none of our specifications is there any evidence that the restrictions reduced the arrest rates for sales or possession of non-narcotics. The estimates for sales of dangerous non-narcotics are fairly precise. The point estimate and standard error for sales of non-narcotics in column 3, which corresponds to the event study, implies a 95 percent confidence interval with a lower end point that rules out reductions bigger than 12 percent. The estimates for possession of non-narcotics are much less precisely estimated, which is unsurprising given the volatility of the event study estimates for that outcome in Figure 10. Similarly, we find no evidence that the laws changed arrest rates for sales or possession of narcotics. While a few of the specifications gave significant point estimates for the sales of addicting narcotics, the findings are not robust across models and the event study shows that the timing of the reduction in the arrest rate is not consistent with the effect being causal.

²¹We also analyzed treatment admissions into publicly funded clinics using the Treatment Episode Data Set. We do not present the results for two reasons. First, we believe our estimates are unreliable because even after controlling for state-specific trends the event study revealed a steady increase in treatment admission rates prior to the enactment of the regulations. This trend continued with approximately the same slope after the OTC restrictions were enacted. Second, given that we find no evidence that the regulations changed methamphetamine consumption there is no reason to expect a change in treatment admissions.

For these reasons we do not interpret our results as evidence that the restrictions changed any of the arrest rates for drug related crimes.

6 Interpretation and Conclusion

The effectiveness of supply-side enforcement efforts to disrupt drug markets and reduce consumption has been widely-debated. The difficulties in estimating the effects of enforcement efforts are widely acknowledged and have compelled researchers to turn to quasi-experimental methods to address some of the limitations of previous research. Legislation to combat the recent methamphetamine expansion provides several natural experiments that can help us understand the benefits of supply-side controls. Still, these efforts have largely relied on time-series variation resulting from federal interventions.

The OTC regulations examined in this paper are the latest in a series of precursor regulations, but variation in the timing of implementation across states provides a unique and important opportunity to estimate the impact of reducing drug use and its harms. In this paper, we link changes in each state’s regulatory environment with changes in the production of methamphetamine. The OTC laws successfully disrupted methamphetamine production with the number of labs declining by 50%. We estimate reductions in laboratories of all sizes, but the largest reduction was amongst the smaller laboratories. The laboratory series fails to show signs of recovery during the first year after the laws were enacted. Reductions in domestic production of methamphetamine represent a gain to society because the process of manufacturing methamphetamine imposes significant harms on the environment.

Despite the large reduction in the production of methamphetamine within a state, we find no evidence of an increase in the purity-adjusted price of methamphetamine or a decline in our measures of methamphetamine usage. A somewhat speculative explanation consistent with our results is that the regulations simply shifted the production of

methamphetamine to outside of the United States while leaving consumption unchanged. This is consistent with DEA evidence that production capacity migrated around the time the state laws were enacted. While it is difficult to precisely estimate the share of methamphetamine that comes from abroad, one study estimates that the share of methamphetamine produced in large labs that are primarily found in Mexico decreased from 80% to 65% just before the state laws were implemented but then reverted back to 80% afterwards.²² The amount of methamphetamine seized at the Southwest border of the United States and the number of laboratories seized in Mexico also increased from 2004 to 2006 by 19 and 35 percent, respectively.²³

Using rich administrative data and quasi-experimental methods that leverage variation in the timing of adoption across states, our analysis suggests that supply-side interventions effectively reduced the number of domestic methamphetamine laboratories. However, we find that domestic methamphetamine consumption was not significantly reduced, which may indicate that the laws resulted in an increase in imported methamphetamine. Unlike other federal precursor regulations, the OTC laws placed a burden on legitimate retailers and consumers of cold medicines. We look forward to characterizing the cost of the regulations on legitimate consumers and retailers of cold medicines in future work.

²²These estimates were taken from the interim report of the Synthetic Drugs Interagency Working Group: https://www.ncjrs.gov/ondcppubs/publications/pdf/interim_rpt.pdf, last accessed 12/31/2012.

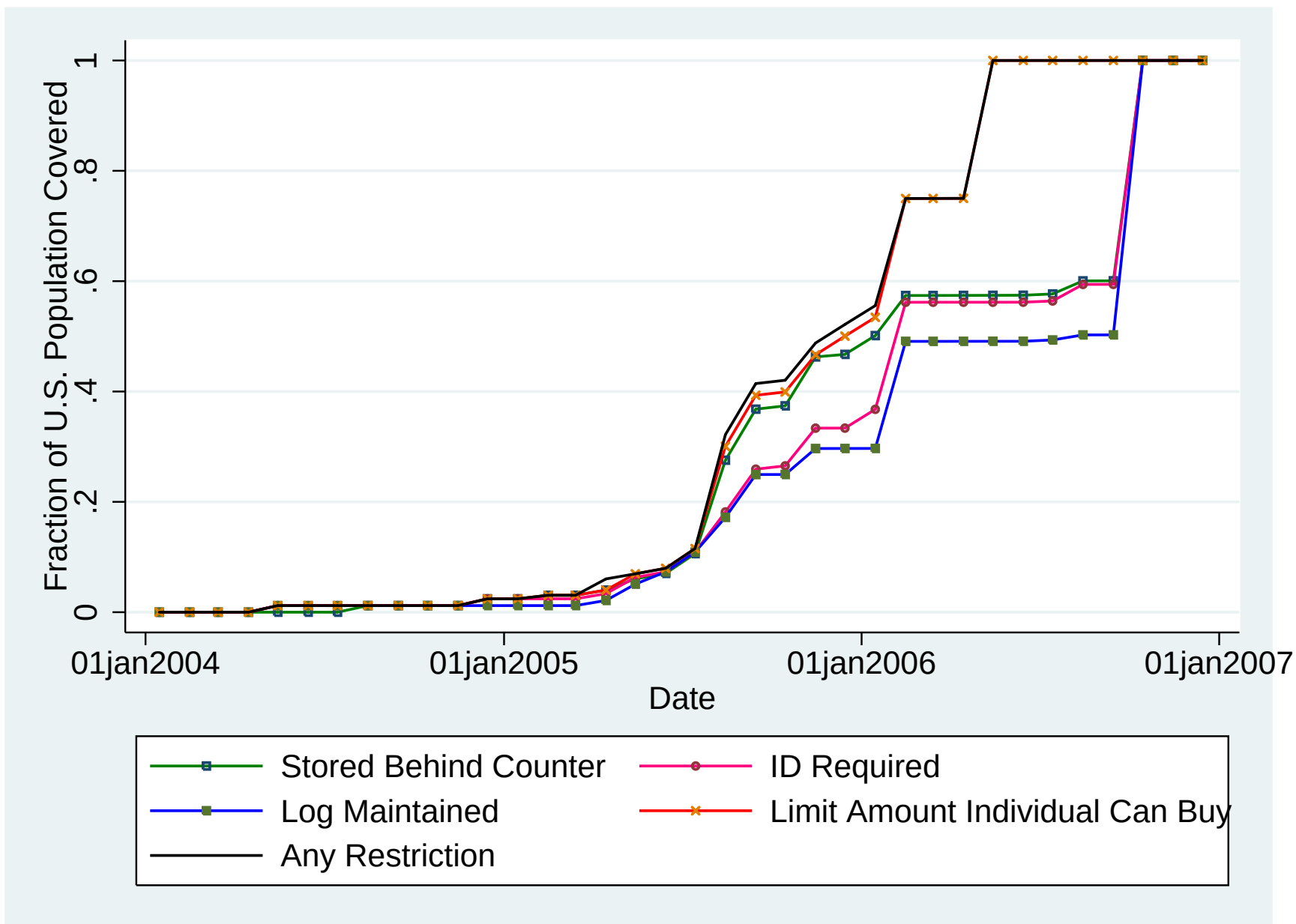
²³These estimates were taken from the Department of Justice's National Drug Intelligence Center National Drug Threat Assessment of 2010, available at <http://www.justice.gov/archive/ndic/pubs38/38661/38661p.pdf>, last accessed 1/2/2013.

References

- CRS**, “Methamphetamine: Background, Prevalence, and Federal Drug Control Policies,” Technical Report, Congressional Research Service (CRS) 2007.
- Cunningham, J.K. and L.M. Liu**, “Impacts of federal ephedrine and pseudoephedrine regulations on methamphetamine-related hospital admissions,” *Addiction*, 2003, *98* (9), 1229–1237.
- DiNardo, J.**, “Law enforcement, the price of cocaine and cocaine use,” *Mathematical and computer Modelling*, 1993, *17* (2), 53–64.
- Dobkin, C. and N. Nicosia**, “The war on drugs: methamphetamine, public health, and crime,” *The American economic review*, 2009, *99* (1), 324.
- Horowitz, J.L.**, “Should the DEA’s STRIDE data be used for economic analyses of markets for illegal drugs?,” *Journal of the American Statistical Association*, 2001, *96* (456), 1254–1271.
- Jacob, Brian, Lars Lefgren, and Enrico Moretti**, “The Dynamics of Criminal Behavior: Evidence from Weather Shocks,” *Journal of Human Resources*, 2007, *42* (3).
- Lynch, M., R. Kemp, L. Krenske, A. Conroy, and J. Webster**, “Patterns of Amphetamine Use: Initial Findings From the Amphetamines in Queensland Research Project, 2003,” *No.: ISBN 1 876986 18 2*, 2003, p. 224.
- Miron, Jeffrey A.**, “The Effect of Drug Prohibition on Drug Prices: Evidence from the Markets for Cocaine and Heroin,” *Review of Economics and Statistics*, 2003, *85* (3), 522–30.

- NIDA**, “Methamphetamine Abuse and Addiction,” Technical Report, National Institute on Drug Abuse 2002.
- ONDCP**, “The Economic Costs of Drug Abuse in the United States, 1999-2002,” Technical Report 2004.
- , “Pushing Back Against Meth: A Progress Report on the Fight Against Methamphetamine in the United States,” Technical Report, Office of National Drug Control Policy 2006.
- , “National Drug Control Budget: FY 2011 Funding Highlights,” Technical Report, Office of National Drug Control Policy February 2010.
- Raphael, Steven and Rudolf Winter-Ember**, “Identifying the Effect of Unemployment on Crime,” *Journal of Law and Economics*, April 2001, 44 (1), 259–83.
- Rawson, R.A., A. Huber, P. Brethen, J. Obert, V. Gulati, S. Shoptaw, and W. Ling**, “Status of methamphetamine users 2–5 years after outpatient treatment,” *Journal of Addictive Diseases*, 2001, 21 (1), 107–119.
- SAMHSA**, “The DASIS Report: Primary Methamphetamine/Amphetamine Admissions to Substance Abuse Treatment: 2005,” 2008.
- Simon, S.L., C.P. Domier, T. Sim, K. Richardson, R.A. Rawson, and W. Ling**, “Cognitive performance of current methamphetamine and cocaine abusers,” *Journal of Addictive Diseases*, 2001, 21 (1), 61–74.
- Weatherburn, D., C. Jones, K. Freeman, and T. Makkai**, “Supply control and harm reduction: lessons from the Australian heroin drought,” *Addiction*, 2002, 98 (1), 83–91.

Yuan, Y. and J.P. Caulkins, “The effect of variation in high-level domestic drug enforcement on variation in drug prices,” *Socio-Economic Planning Sciences*, 1998, 32 (4), 265–276.



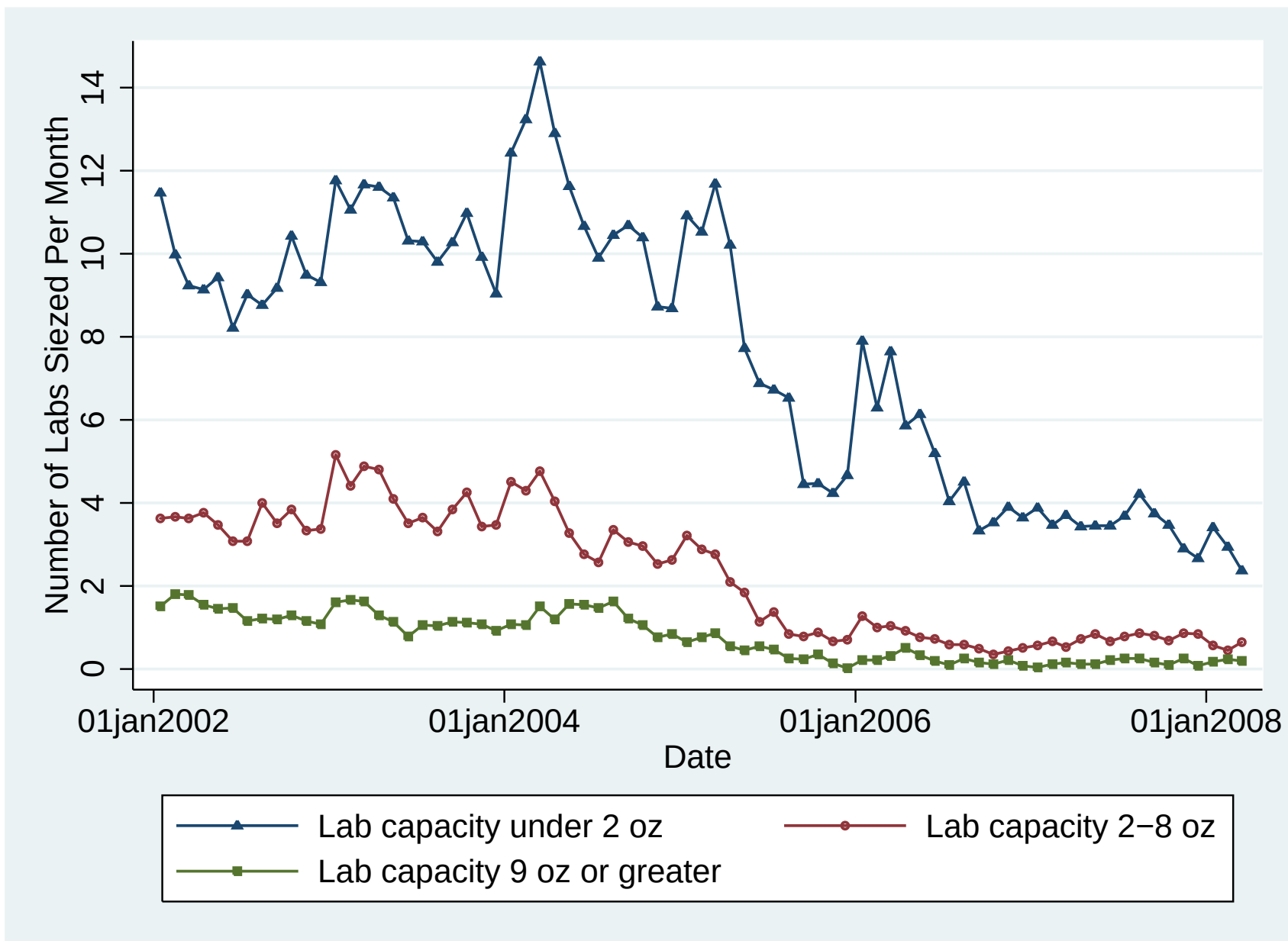


Figure 2: Methamphetamine Labs Discovered or Seized by Capacity

Notes: The time series above are constructed using records from the National Clandestine Laboratory Seizure System. The figure contains the average number of labs discovered in a state by month.

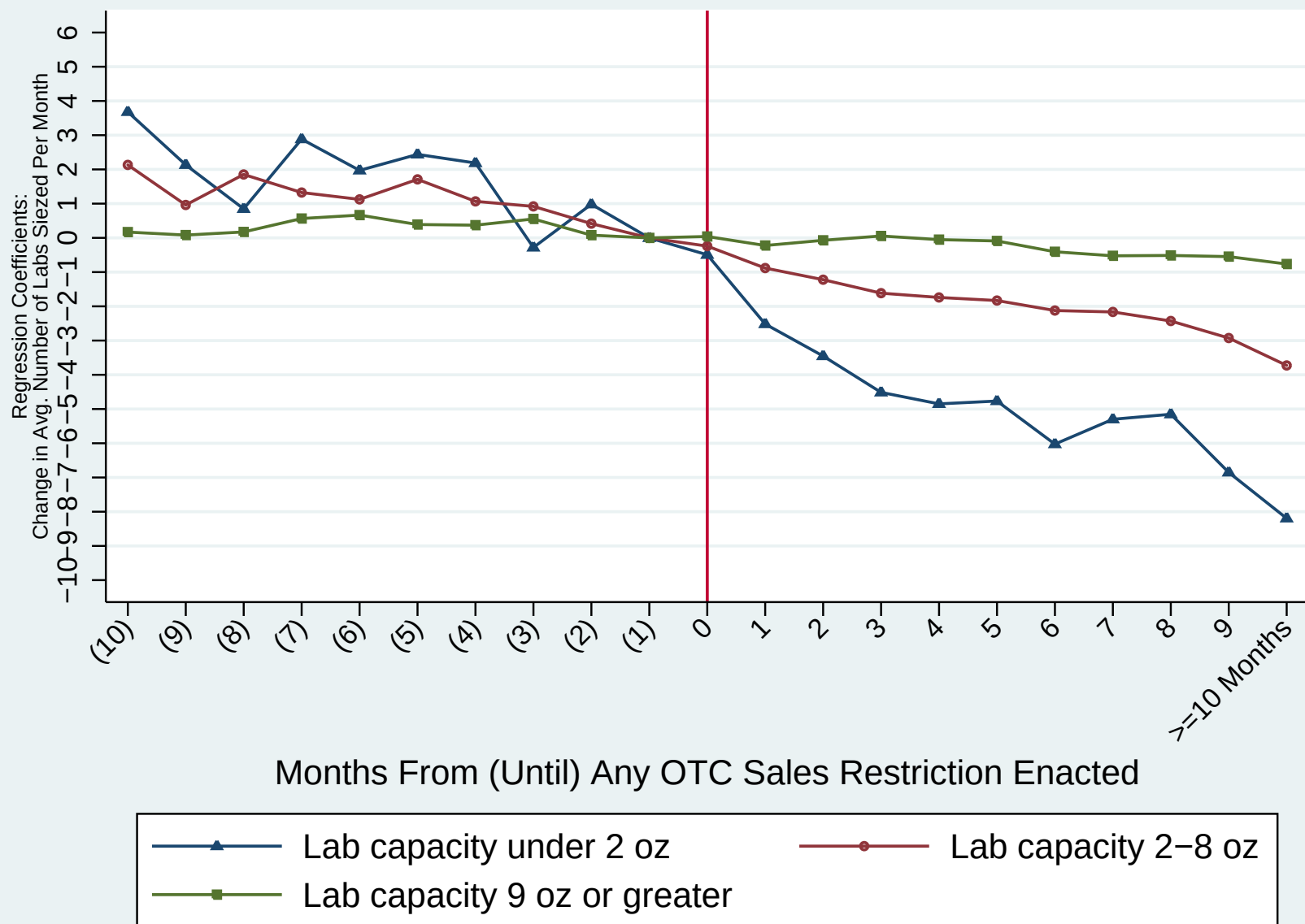


Figure 3: Event Study: Methamphetamine Labs Discovered or Seized by Capacity

Notes: Count of labs discovered was regressed on state fixed effects, year/month fixed effects, linear and quadratic state time trends, and indicators corresponding to the number of months since any over the counter restriction went into effect. The graph contains OLS estimates of the coefficients on the indicators corresponding to the number of months since any over the counter restriction went into effect. The coefficient on the event dummy equal to one if a state enacted any OTC restriction in the next month was normalized to zero. The estimates include records for all 50 states and the District of Columbia from January 2002 through March 2008.

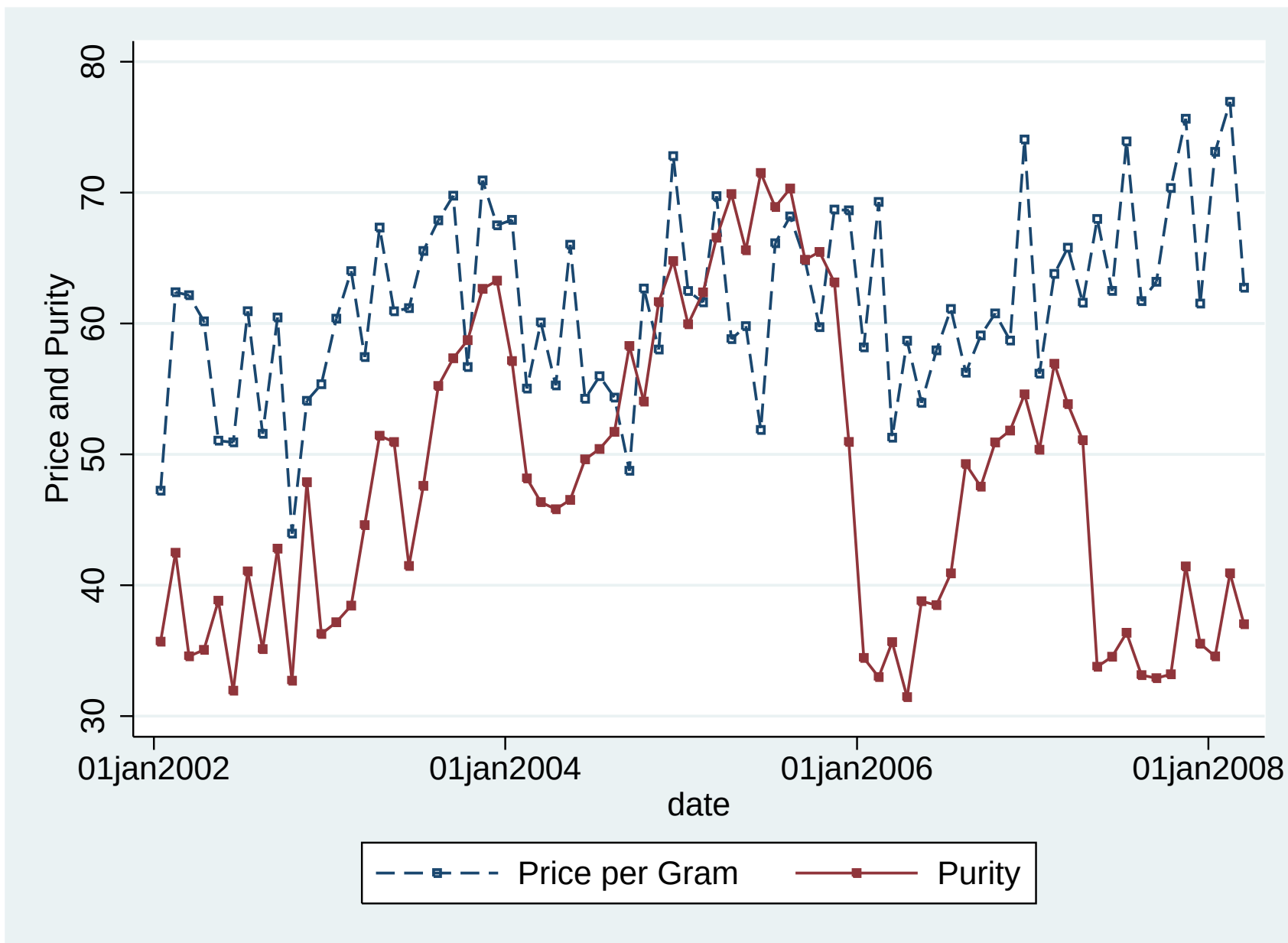


Figure 4: Price and Purity of Methamphetamine from STRIDE

Notes: Price is measured in dollars and purity is measured in percent. The time series contains averages of the average price per gram and purity of drugs purchased by the police in a state by month.

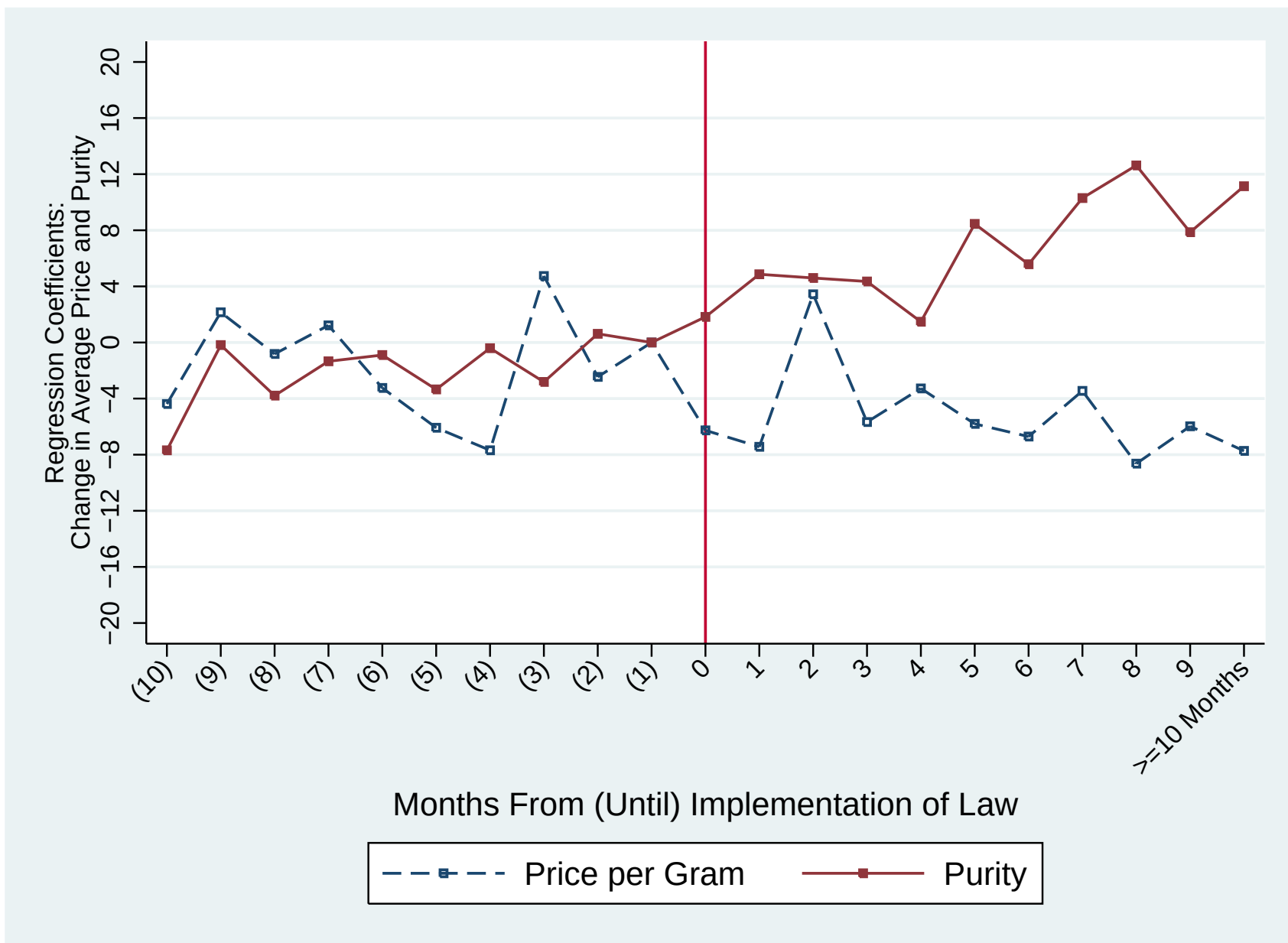


Figure 5: Event Study: Price and Purity of Methamphetamine from STRIDE

Notes: Price is measured in dollars and purity is measured in percent. Average Price and Average Purity were regressed on state fixed effects, year/month fixed effects, linear and quadratic state time trends, and indicators corresponding to the number of months since any over the counter restriction went into effect. The regressions were weighted by the number of illegal drug transactions used to derive the average price and purity measures in a state/month. The graph contains OLS estimates of the coefficients on the indicators corresponding to the number of months since any over the counter restriction went into effect. The coefficient on the event dummy equal to one if a state passed an over counter restriction in the next month was normalized to zero. The estimates include records from the District of Columbia and all 50 states except for Nebraska for January 2002 through March 2008. In many smaller states there are months without any purchases.

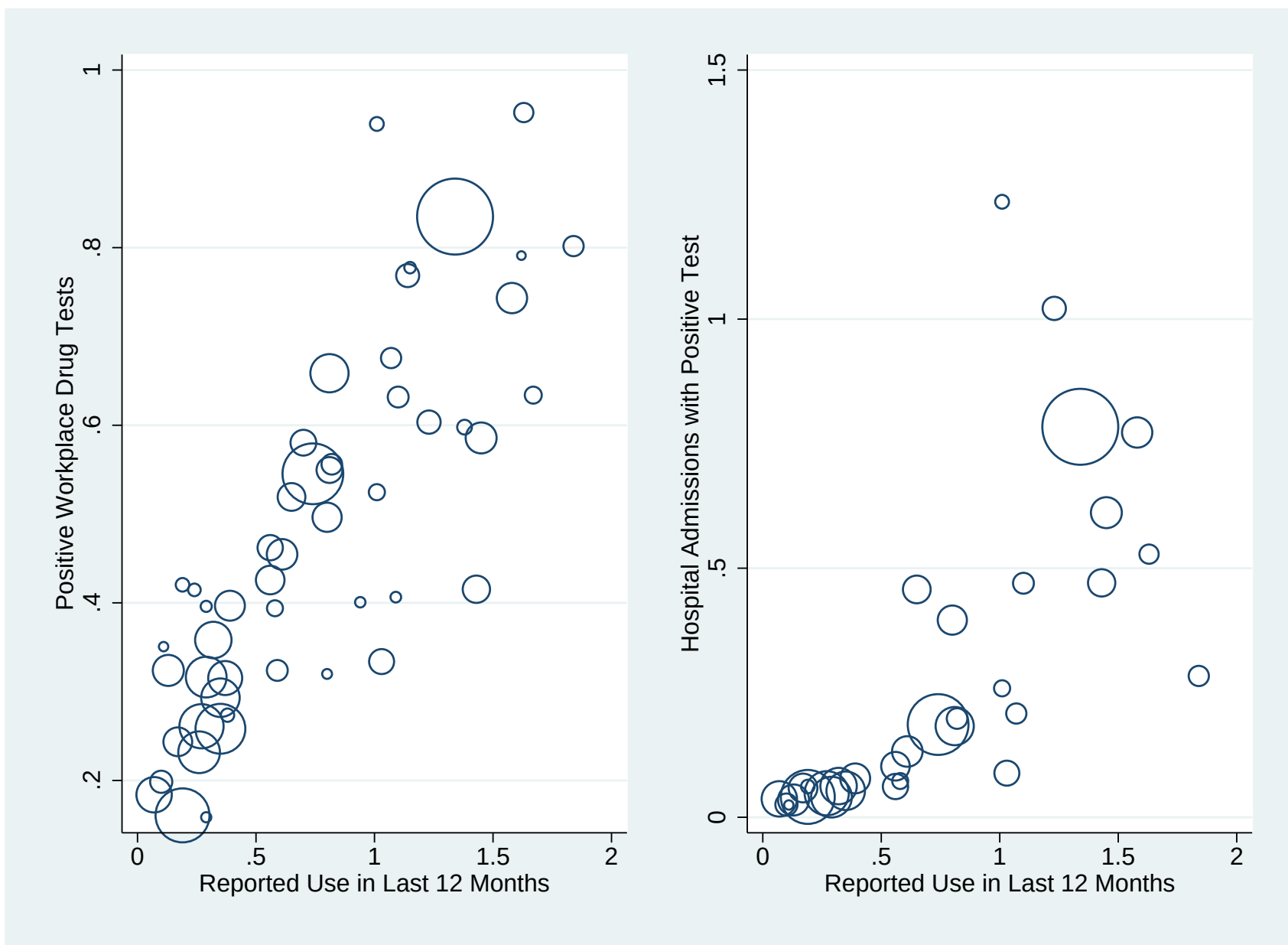


Figure 6: Correlations of Measures of Methamphetamine Use

Notes: The figures are scatter plots of the average percent of drug tests that are positive against the average percent of survey respondents in the National Survey on Drug Use and Health that report using methamphetamine in the past 12 months. The averages are calculated during the period from 2004 to 2007. Each bubble represents a state and the size of each bubble is proportional to state population.

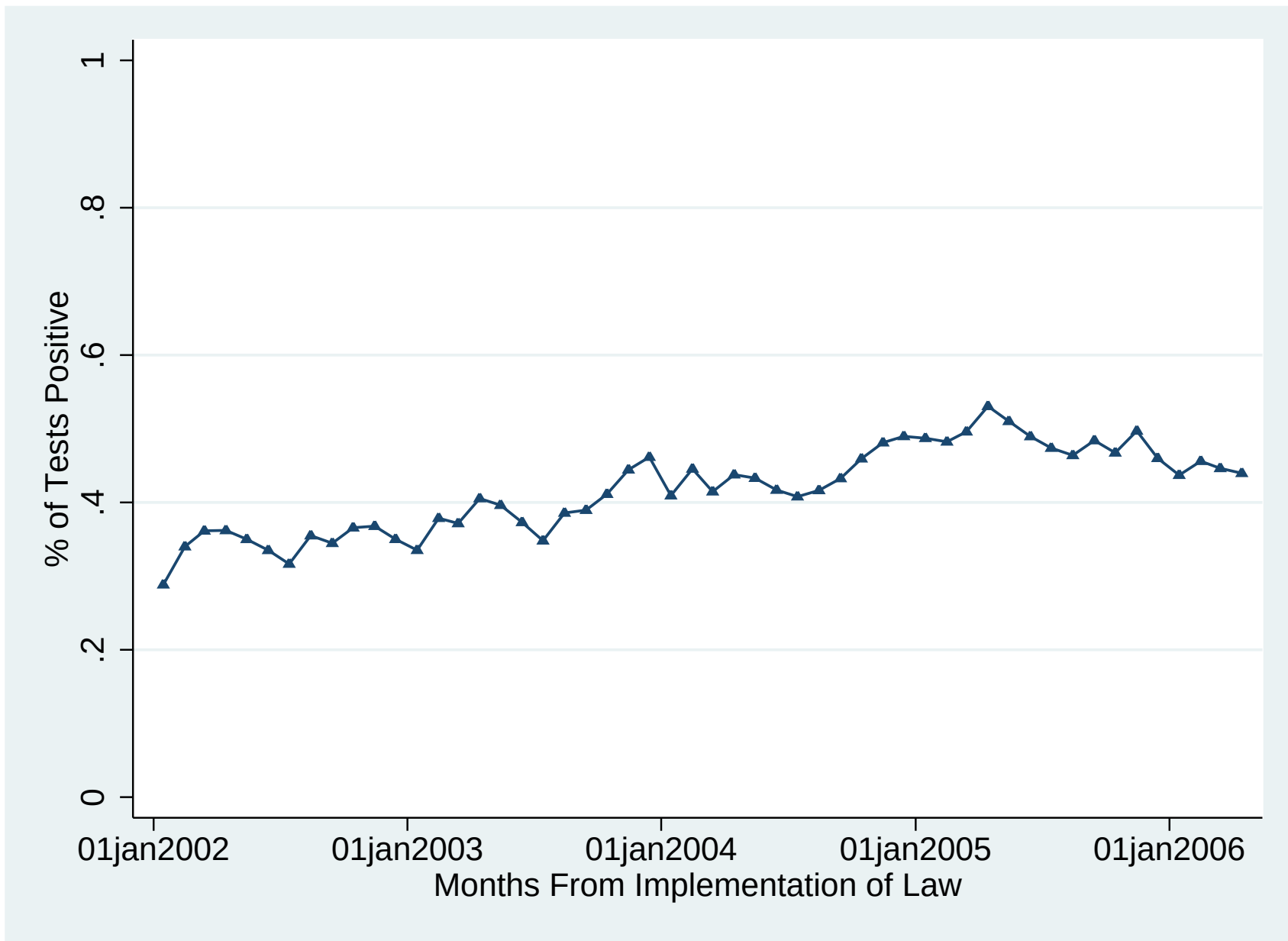


Figure 5: Percent of Workplace Drug Tests that are Positive for Amphetamines

Notes: The time series contains the average percent of positive workplace drug tests in a state by month. The data used to construct the time series above are extracted from the Office of National Drug Control Policy report “Pushing Back Against Meth”. It was downloaded on February 2, 2007 from www.whitehousedrugpolicy.gov/publications/pdf/pushingback_against_meth.pdf. The time series is weighted by state population. The unweighted time series has a similar shape but is more variable.

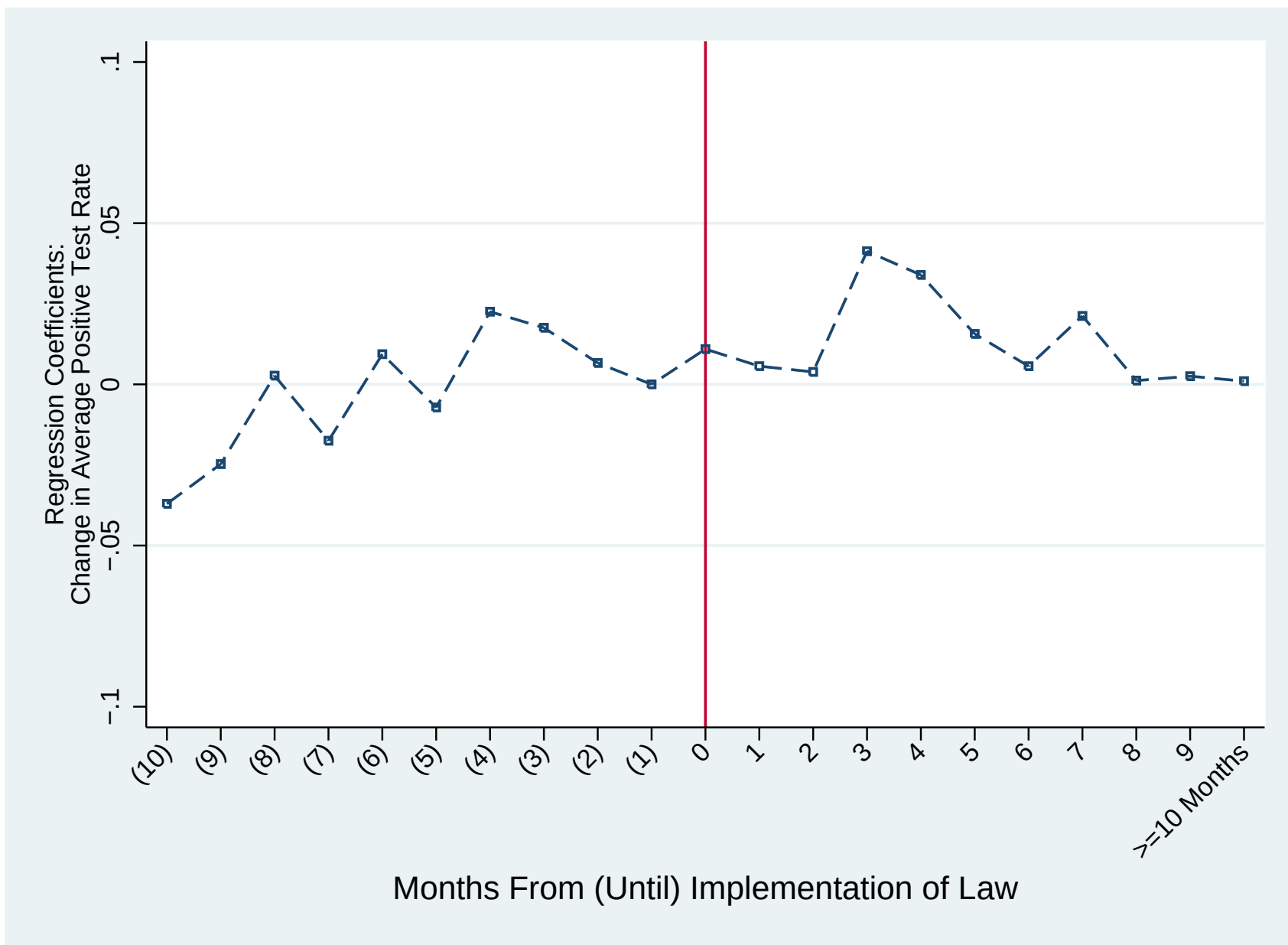


Figure 6: Event Study: Percent of Workplace Drug Tests that are Positive for Amphetamines

Notes: Percent of positive tests was regressed on state dummies, year/month fixed effects, state time trends, and indicators corresponding to the number of months since any over the counter restriction went into effect. The regressions were weighted by state population. The graph contains OLS estimates of the coefficients on the indicators corresponding to the number of months since any over the counter restriction went into effect. The coefficient on the event dummy equal to one if a state passed an over counter restriction in the next month was normalized to zero. The data used to construct the graph above are extracted from the Office of National Drug Control Policy report “Pushing Back Against Meth”. It was downloaded on February 2, 2007 from www.whitehousedrugpolicy.gov/publications/pdf/pushingback_against_meth.pdf. The estimates include records from the District of Columbia and all 50 states for January 2000 through April 2006.

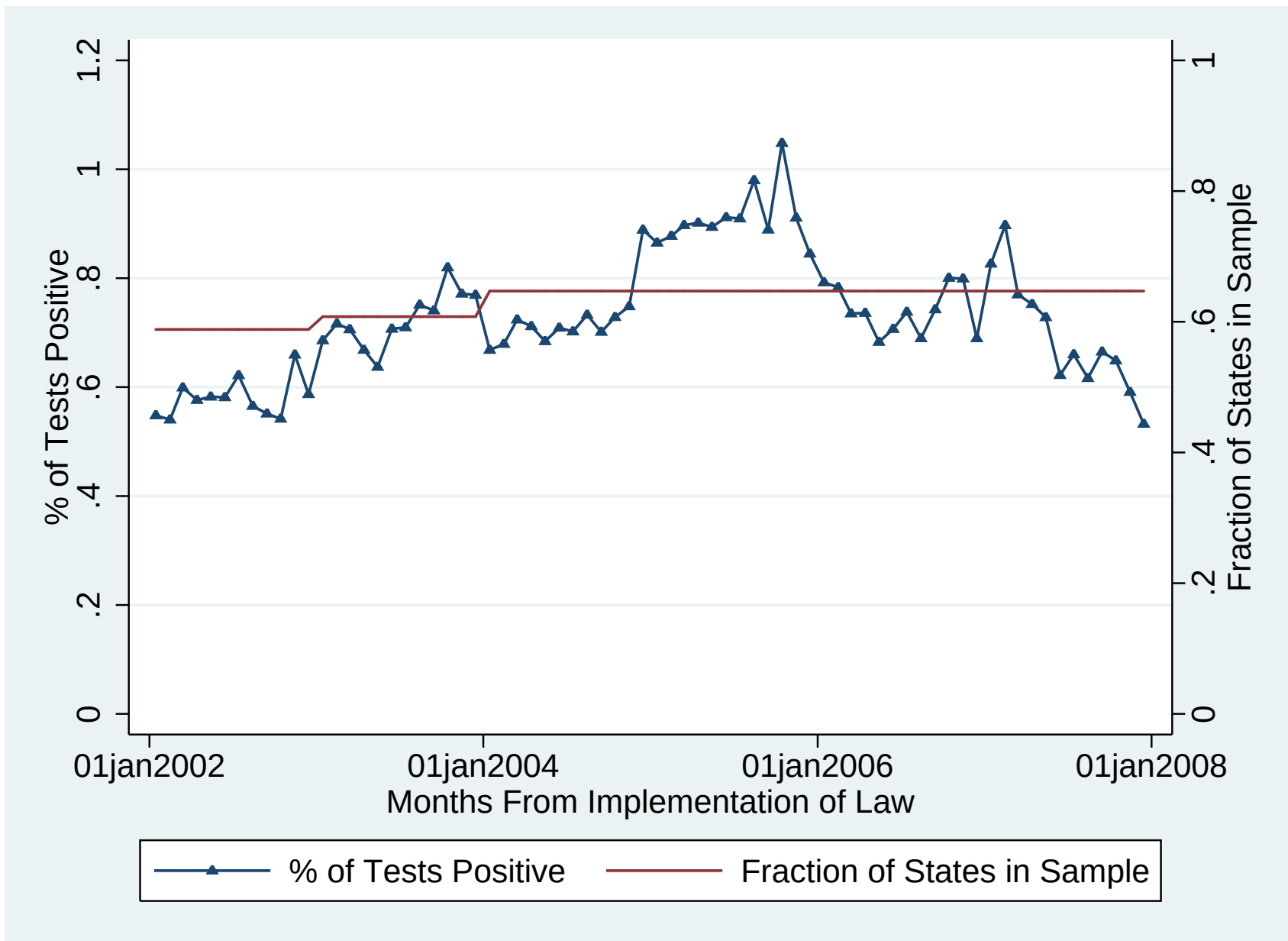


Figure 7: Percent of Hospital Admissions with Positive Drug Test for Amphetamine (Ages 15-40)

Notes: This series is derived from the HCUP NIS which includes a 20 percent sample of community hospitals from the following states AR, CA, CO, CT, GA, HI, IL, IN, KS, KY, MA, MD, MI, MN, MO, NC, NE, NH, NV, NY, OH, OR, SC, TN, TX, UT, VT, WI, WV and 100 percent of the community hospitals from AZ, NJ, and WA. For Iowa it is a 20 percent sample before 2004 and all community hospitals from 2004-2007. Approximately 90 percent of hospital stays occur at community hospitals.



Figure 8: Event Study: Percent of Hospital Admissions with Positive Drug Test for Amphetamine (Ages 15-40)

Notes: Percent of hospitalizations among 15-40 year olds that test positive for amphetamines was regressed on state fixed effects, year/month fixed effects, state time trends, and indicators corresponding to the number of months since any over the counter restriction went into effect. The regressions were weighted by state population between the ages of 15 and 40. The graph contains OLS estimates of the coefficients on the indicators corresponding to the number of months since any over the counter restriction went into effect. The coefficient on the dummy variable equal to one if a state passed an over counter restriction in the next month was normalized to zero. This series is derived from the HCUP NIS which includes a 20 percent sample of community hospitals from the following states AR, CA, CO, CT, GA, HI, IL, IN, KS, KY, MA, MD, MI, MN, MO, NC, NE, NH, NV, NY, OH, OR, SC, TN, TX, UT, VT, WI, WV and 100 percent of the community hospitals from AZ, NJ, and WA. For Iowa it is a 20 percent sample before 2004 and all community hospitals from 2004-2007. Approximately 90 percent of hospital stays occur at community hospitals. The estimates include records for January 2002 through December 2007.

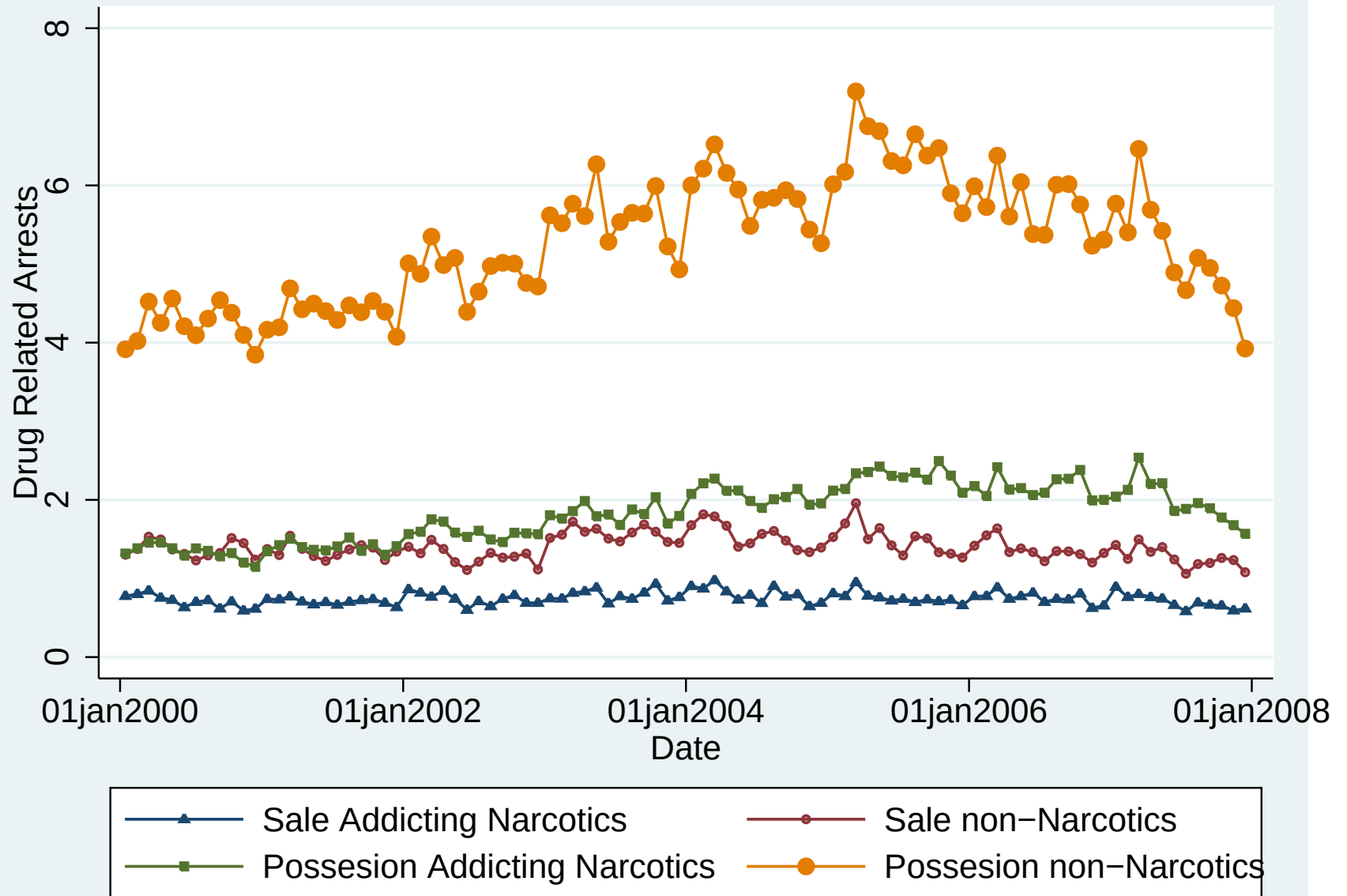


Figure 9: Drug Related Arrest Rate from Uniform Crime Reports

Notes: The series above are derived from the Uniform Crime Reports for January 2002 through December 2007. There is no data available for Alabama, Florida, and Rhode Island. Records from agencies in some states that report either annually or biannually have been dropped.

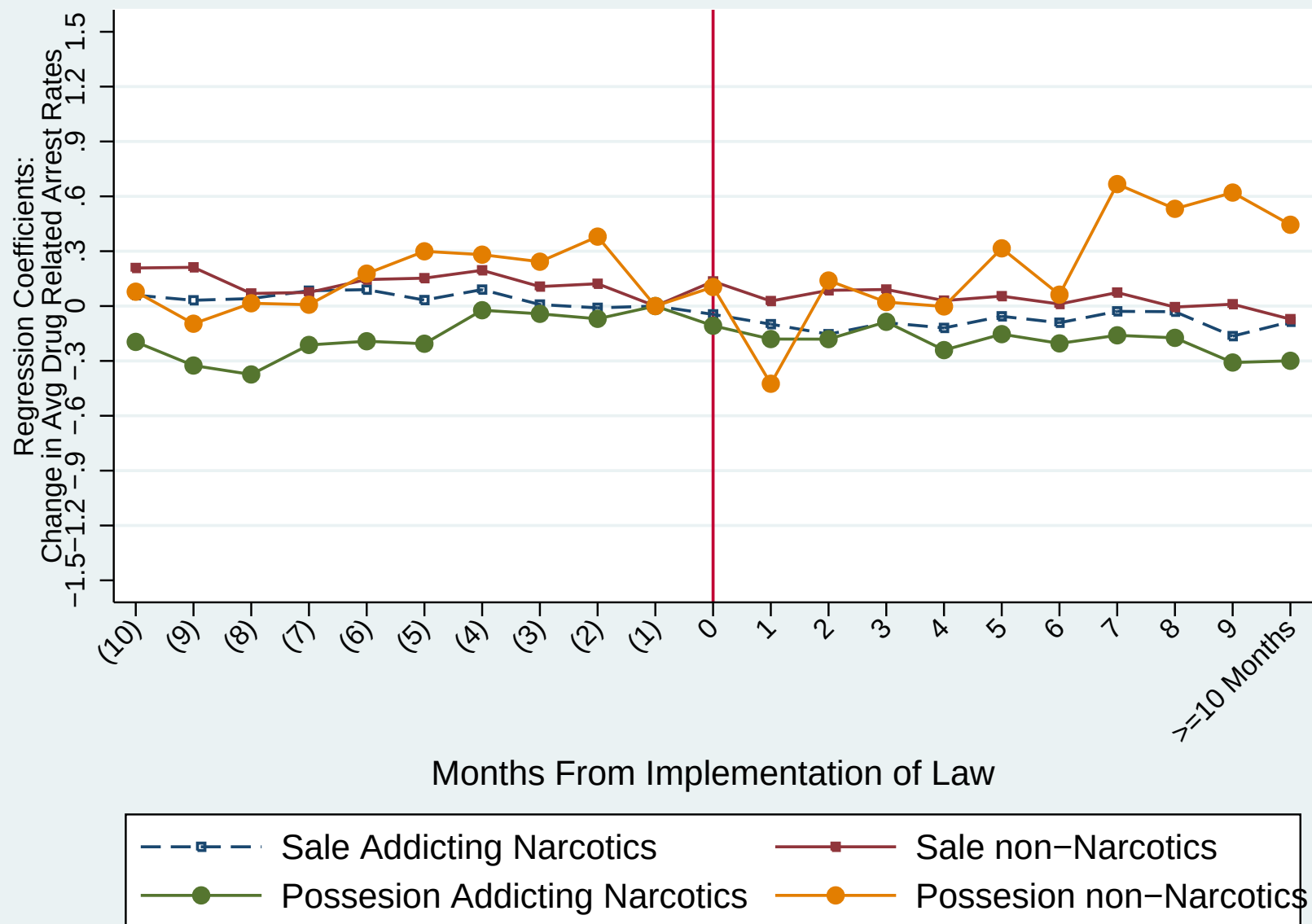


Figure 10: Event Study: Drug Related Arrest Rate from Uniform Crime Reports

Notes: Drug related arrest rates in a month per 10,000 people were regressed on state fixed effects, calendar time effects, state time trends, and indicators corresponding to the number of months since any over the counter restriction went into effect. The regression was weighted by state population. The graph contains OLS estimates of the coefficients on the indicators corresponding to the number of months since any over the counter restriction went into effect. The coefficient on the dummy variable equal to one if a state passed an over counter restriction in the next month was normalized to zero. The series above are derived from the Uniform Crime Reports for January 2002 through December 2007. There is no data available for Alabama, Florida, and Rhode Island. Records from agencies in some states that report either annually or biannually have been dropped.

Table 1: State Laws Regulating the Sales of Over the Counter Medicines Containing Ephedrine or Pseudoephedrine

			<u>Correlations</u>			
	Limit on Amount	ID Required	Log Maintained	Stored Behind Counter	Any Restriction	States Adopting Before Federal Law
Limit on Amount an Individual Can Purchase	1					35
ID Required at Time of Purchase	0.90	1				30
Log Maintained of Purchaser's Identity	0.88	0.97	1			24
Stored Behind Counter, in Line Of Sight or on Video	0.92	0.91	0.88	1		35
Any of the Restrictions Above	1.00	0.90	0.87	0.92	1	35

Notes: The correlations are all estimated weighted for state populations on data from January of 2004 through March of 2008. Only two states of the 35 states that implemented product placement restrictions (Louisiana and Michigan) allowed the items to be monitored by video or placed in a clerk's line of sight rather than behind a counter or in a locked counter.

Table 2: Impact of OTC Regulations on Methamphetamine Lab Seizures

	<u>Number of Labs Seized</u>				<u>Lab Capacity Under 2 Ounces</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	-6.71	-5.05	-7.02	-7.30	-4.08	-3.05	-4.73	-4.92
	(1.52)	(1.44)	(1.87)	(1.93)	(0.90)	(0.89)	(1.31)	(1.36)
Mean Prior to OTC Restriction	13.94	13.94	13.94	14.46	9.59	9.59	9.59	9.93
Observations	3825	3825	3825	3675	3825	3825	3825	3675
Number of States	51	51	51	49	51	51	51	49
	<u>Lab Capacity 2 to 8 Ounces</u>				<u>Lab Capacity 9 Ounces or More</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	-2.29	-1.76	-1.87	-1.94	-0.34	-0.24	-0.42	-0.44
	(0.55)	(0.47)	(0.56)	(0.58)	(0.22)	(0.19)	(0.21)	(0.21)
Mean Prior to OTC Restriction	3.25	3.25	3.25	3.38	1.10	1.10	1.10	1.15
Observations	3825	3825	3825	3675	3825	3825	3825	3675
Number of States	51	51	51	49	51	51	51	49
Linear State Trends	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Quadratic State Trends	No	No	Yes	Yes	No	No	Yes	Yes
Covariates	No	No	No	Yes	No	No	No	Yes

Notes: All regressions include state fixed effects and year/month fixed effects. The dependent variable in the regressions is count of labs seized or discovered in a month in a particular state. These are derived from the National Clandestine Laboratory Seizure System. The estimates include records for all 50 states and the District of Columbia from January 2002 through March 2008 $[(6*12+3)*51=3825]$. Covariates include the unemployment rate, number of households receiving food stamps, average temperature and precipitation. In the third specification we drop 150 observations because weather data is missing for Alaska and Hawaii.

Table 3: Impact of OTC Regulations on Drug Price and Purity

	<u>Price per Gram</u>				<u>Purity</u>			
	<u>Methamphetamine</u>				<u>Methamphetamine</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	2.89	2.19	-1.83	-2.12	6.15	4.49	3.81	4.18
	(2.94)	(2.93)	(3.54)	(3.57)	(2.58)	(2.55)	(2.24)	(2.26)
Mean Prior to OTC Restriction	49.13	49.13	49.13	48.60	56.64	56.64	56.64	56.32
Observations	2,074	2,074	2,074	1,995	2,074	2,074	2,074	1,995
Number of States	49	49	49	47	49	49	49	47
	<u>Price per Gram</u>				<u>Purity</u>			
	<u>Cocaine</u>				<u>Cocaine</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	1.43	1.46	2.18	2.08	0.023	0.020	0.14	0.082
	(2.37)	(2.38)	(2.58)	(2.64)	(1.29)	(1.52)	(1.65)	(1.70)
Mean Prior to OTC Restriction	40.66	40.66	40.66	61.26	40.64	40.64	40.64	61.30
Observations	2,449	2,449	2,449	2,423	2,449	2,449	2,449	2,423
Number of States	50	50	50	48	50	50	50	48
	<u>Price per Gram</u>				<u>Purity</u>			
	<u>Crack</u>				<u>Crack</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	-3.94	-3.19	-1.39	-1.52	0.19	0.19	-0.42	-0.50
	(2.68)	(2.33)	(3.36)	(3.21)	(1.36)	(1.28)	(1.14)	(1.18)
Mean Prior to OTC Restriction	62.69	62.69	62.69	62.68	63.55	63.55	63.55	63.19
Observations	2,381	2,381	2,381	2,342	2,381	2,381	2,381	2,342
Number of States	50	50	50	48	50	50	50	48
	<u>Price per Gram</u>				<u>Purity</u>			
	<u>Heroin</u>				<u>Heroin</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	-6.41	-6.44	-4.95	-3.81	3.51	2.40	3.00	3.37
	(12.2)	(11.5)	(11.1)	(11.2)	(2.65)	(2.61)	(3.05)	(3.02)
Mean Prior to OTC Restriction	122.04	122.04	122.04	122.03	45.79	45.79	45.79	45.77
Observations	1,166	1,166	1,166	1,165	1,166	1,166	1,166	1,165
Number of States	44	44	44	43	44	44	44	43
Linear State Time Trends	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Quadratic State Time Trends	No	No	Yes	Yes	No	No	Yes	Yes
Covariates	No	No	No	Yes	No	No	No	Yes

Notes: All regressions include state fixed effects and year/month fixed effects. Standard errors are clustered at the state level. The dependent value in the regressions is average price and average purity over the month of drugs purchased by law enforcement. Price is measured in dollars and purity is measured in percent. These are derived from the National Clandestine Laboratory Seizure System. The estimates include records from the District of Columbia and all 50 states except for Nebraska for January 2002 through March 2008 $[(6*12+3)*50=3750]$. In many smaller states there are months without any purchases particularly for Heroin and Methamphetamine. Covariates include the unemployment rate, number of households receiving food stamps, average temperature and precipitation. In the third specification we drop 150 observations because weather data is missing for Alaska and Hawaii.

Table 4: Percent of Workplace Drug Tests Positive for Methamphetamine

	<u>Percent of Tests Positive</u>			
	(1)	(2)	(3)	(4)
OTC Restriction	0.031 (0.018)	-0.012 -0.017	-0.0045 (0.018)	-0.0072 (0.018)
Mean Prior to OTC Restriction	0.365	0.365	0.365	0.362
Observations	3,876	3,876	3,876	3,724
Number of States	51	51	51	49
Linear State Time Trends	No	Yes	Yes	Yes
Quadratic State Time Trends	No	No	Yes	Yes
Covariates	No	No	No	Yes

Notes: All regressions include state fixed effects and year/month fixed effects and were weighted by state population. Standard errors are clustered at the state level. The data used to construct the dependent variable are extracted from the Office of National Drug Control Policy report "Pushing Back Against Meth". It was downloaded on February 2, 2007 from www.whitehousedrugpolicy.gov/publications/pdf/pushingback_against_meth.pdf. The dependent variable is the percent of drug tests that are positive. The estimates include records from the District of Columbia and all 50 states for January 2000 through April 2006 [(6*12+4)*51=3876]. Covariates include the unemployment rate, number of households receiving food stamps, average temperature and precipitation.

Table 5: Percent of Hospitalizations with a Positive Drug Test

	<u>Amphetamine</u>				<u>Opioids</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	0.11 (0.093)	0.083 (0.091)	0.060 (0.057)	0.074 (0.055)	-0.014 (0.15)	-0.093 (0.13)	-0.077 (0.12)	-0.11 (0.15)
Mean Prior to OTC Restriction	0.671	0.671	0.671	0.656	1.912	1.912	1.912	1.928
Observations	2316	2316	2316	2244	2316	2316	2316	2244
Number of States	33	33	33	32	33	33	33	32
	<u>Cocaine</u>				<u>Marijuana</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	-0.33 (0.20)	-0.29 (0.19)	-0.18 (0.16)	-0.22 (0.18)	-0.035 (0.13)	0.0041 (0.12)	0.11 (0.13)	0.11 (0.14)
Mean Prior to OTC Restriction	2.201	2.201	2.201	2.201	1.895	1.895	1.895	1.896
Observations	2316	2316	2316	2244	2316	2316	2316	2244
Number of States	33	33	33	32	33	33	33	32
Linear State Time Trends	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Quadratic State Time Trends	No	Yes	Yes	Yes	No	No	Yes	Yes
Covariates	No	No	No	Yes	No	No	No	Yes

Notes: All specifications include state fixed effects and year/month fixed effects. Regressions are weighted by population between the ages of 15 and 40. Standard errors are clustered by state. The dependent variable in the regressions is the percent of hospitalizations among 15 to 40 years olds in a month in a state that test positive for a particular drug. This dependent variable is derived from the HCUP NIS which includes a 20 percent sample of community hospitals from the following states AR, CA, CO, CT, GA, HI, IL, IN, KS, KY, MA, MD, MI, MN, MO, NC, NE, NH, NV, NY, OH, OK, OR, SC, TN, TX, UT, VT, WI, WV and 100 percent of the community hospitals from AZ, NJ and WA. For Iowa it is a 20 percent sample before 2004 and all community hospitals from 2004-2007. Approximately 90 percent of hospital visits occur at community hospitals. The hospital records do not distinguish positive tests from methamphetamine from positive tests from amphetamine. In this period over 90 percent of positive tests for either methamphetamine or amphetamine are due to methamphetamine. The estimates include records for January 2002 through December 2007.

Table 6: Impact of OTC Regulations on Drug Related Arrests

	<u>Sales of Addicting Narcotics</u>				<u>Sales of Dangerous non-Narcotics</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	-0.053 (0.057)	-0.061 (0.042)	-0.13 (0.042)	-0.13 (0.043)	0.012 (0.073)	0.026 (0.071)	-0.087 (0.063)	-0.091 (0.067)
Rate per 10,000	0.726	0.726	0.726	0.728	1.746	1.746	1.746	1.577
Observations	3,477	3,477	3,477	3,327	3,477	3,477	3,477	3,327
Number of States	48	48	48	46	48	48	48	46
	<u>Possession of Addicting Narcotics</u>				<u>Possession of Dangerous non-Narcotics</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	0.17 (0.11)	0.16 (0.096)	-0.038 (0.091)	-0.025 (0.10)	-0.11 (0.74)	-0.12 (0.76)	-0.24 (0.37)	-0.23 (0.35)
Rate per 10,000	1.572	1.572	1.572	1.577	8.409	8.409	8.409	8.428
Observations	3,477	3,477	3,477	3,327	3,477	3,477	3,477	3,327
Number of States	48	48	48	46	48	48	48	46
Linear State Time Trends	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Quadratic State Time Trends	No	No	Yes	Yes	No	No	Yes	Yes
Covariates	No	No	No	Yes	No	No	No	Yes

Notes: All regressions include state fixed effects and month dummies. Standard errors are clustered at the state level. The dependent value in the regressions is arrest rate in a month per 10,000 people in a state. Regressions are weighted by state population. These are derived from the Uniform Crime Reports for January 2002 through December 2007. There is no data available for Alabama, Florida, and Rhode Island. Records from agencies in some states that report either annually or biannually have been dropped.